

Problem ID SIH26034

Department DoCA

Tests 562 Passed

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AI-Assisted Legal Metrology Inspection & Evidence Workstation

Statutory Compliance Verification & Evidentiary Dossier Generation for the Department of Consumer Affairs (DoCA)

📷 75 Packaging Photos • 📦 7 Golden Demo SKUs • 🖥️ Offline Field Mode
Evidence-Linked Traceability • Human-Reviewed Adjudication • Section 63 BSA
Support

🐍 Python 3.11+

🚀 FastAPI 0.110+

⚙️ React 18

🔧 ONNX Runtime CPU

Overview

Nirikshak is a digital inspection workstation for Legal Metrology officers.

It turns fragmented packaging inspection into one traceable workflow: **capture evidence, measure, extract declarations, check rules, review findings, and generate the inspection dossier.**

The Core Differentiator

Evidence-linked inspection with automated analysis, physical measurement, deterministic statutory checks, and mandatory officer adjudication in one workflow.

Every finding remains traceable to the evidence that produced it:

 Nirikshak evidence traceability from finding to preserved dossier

Nirikshak incorporates workflow safeguards that keep automated analysis subordinate to authorized officer review. The system serves strictly as an **Augmented Diagnostic Assistant**—it never issues compounding orders, penalty notices, or citations autonomously. Every statutory adjudication remains under the explicit control of a qualified Legal Metrology Officer (LMO).

Problem & Solution

The Problem (SIH26034)

Enforcement officers inspecting packaged commodities in wholesale mandis, retail outlets, and e-commerce fulfillment hubs face persistent operational hurdles:

1. **Scattered Evidence & Broken Traceability:** Officers capture loose photos on personal phones, take manual notes in paper registers, and calculate metrics by hand—creating fractured chains of custody vulnerable to evidentiary challenge.
2. **Cognitive Load Across Multilingual Packaging:** Transcribing mandatory statutory declarations (MRP, Net Quantity, Unit Sale Price, Manufacturing Date, Consumer Care, Country of Origin) across English and regional Indic scripts is labor-intensive and prone to error.
3. **Complex Metric Font Verification:** Verifying minimum numeral font heights across packaging surface areas under the Legal Metrology (Packaged Commodities) Rules, 2011 (varying from 1.0 mm to 6.0 mm) requires physical calipers and manual geometric calculations that are difficult to standardize in field conditions.
4. **Black-Box Automation Pitfalls:** Automated scanners often output opaque, unverifiable verdicts without showing underlying coordinate evidence or permitting human correction, failing administrative standards.

The Solution: Nirikshak

Nirikshak unifies field intake, calibrated computer vision, multilingual scene text recognition, deterministic statutory logic, and human adjudication into a single web platform:

 How Nirikshak transforms fragmented packaging inspection into one traceable workflow

Flagship Inspector Workflow

The complete inspection flow:

 Nirikshak flagship inspector workflow

The key idea: Every automated finding remains linked to the evidence that produced it, and the authorized officer makes the final disposition.

For complete pipeline stage specifications, sequence diagrams, and mathematical formulations, see [docs/specifications/o3_FINAL_ARCHITECTURE.md](https://docs.specifications/o3_FINAL_ARCHITECTURE.md).

Human-in-the-Loop Adjudication

Nirikshak places the human officer at the center of all enforcement decisions:

 Nirikshak human-in-the-loop adjudication canvas

- **AI findings stay diagnostic:** automated analysis never becomes the final legal disposition.
- **Every finding is inspectable:** officers can trace it to the source evidence, extracted field and governing rule.
- **Overrides are accountable:** changes require written justification and remain in the audit trail.

Administrative authority is strictly separated via role-based access control (RBAC): field inspectors propose preliminary recommendations, while authorized Controllers review and sign formal statutory notices.

Key Capabilities

1. Evidence Intake & Optical Quality Gate

- **Guided Viewfinder HUD:** Real-time optical assistance prompts field officers for camera stabilization, framing distance, and glare control.
- **Degradation Screening:** Automatically detects excessive blur and specular glare, routing unreadable inputs to guided recapture.
- **Multi-Source Ingestion:** Supports live camera captures, high-resolution multi-facet photos, and Rule 6(10) e-commerce URLs.

2. Physical Measurement & Metric Calibration

- **Scale Calibration:** Computes physical millimeter scale from standard ISO-7810 debit/credit cards ($85.60\text{ mm} \times 53.98\text{ mm}$) or ArUco markers (50.0 mm) on standard smartphone/camera hardware.
- **PDP Surface Area:** Determines physical packaging face area across rectangular, cylindrical, and flexible packages.
- **Numeral Font Measurement:** Measures physical millimeter numeral height to verify compliance against statutory minimums.

3. Multilingual Declaration Extraction & Vision Probing

- **Bilingual Scene Text:** Transcribes packaging declarations across English and Devanagari Hindi text.
- **180° Inversion Auto-Detection:** Automatically probes inverted 180° orientation on low-confidence crops ($p_{\text{conf}} < 0.92$), transcribing upside-down physical packaging labels with > 0.98 confidence.
- **Multi-Facet Cross-Panel Fusion:** Fuses declarations across multiple package panels (FRONT_PDP , SIDE_PANEL , MACRO_CLOSE_UP , STAMP , BOTTOM_BASE , BACK_PANEL) into a unified statutory record.
- **Statutory Entity Parsing:** Extracts mandatory declarations: MRP, Net Quantity, Unit Sale Price, Dates, Consumer Care, and Origin.

- **Prohibited Unit Detection:** Flags non-standard metric abbreviations (`gms` , `gm` , `ML` , `ltrs`) prohibited under Section 11.

4. Deterministic Compliance Analysis

- **Table-I Font Schedules:** Verifies numeral height against surface-area statutory schedules without probabilistic drift.
- **USP Arithmetic Consistency:** Checks Unit Sale Price calculations against declared MRP and Net Quantity.
- **Statutory Rules & Epochs:** Evaluates declarations against packaging date amendments and Rule 6(10) digital marketplace exemptions.

5. Evidence-Linked Officer Adjudication

- **Interactive Split Canvas:** Side-by-side packaging photo inspection with vector bounding polygons and statutory findings.
- **Audit-Logged Overrides:** Requires mandatory written justification remarks whenever an officer overrides a recommendation.
- **Case State Lifecycle:** Enforces structured case progression from initial registration to final administrative disposition.

6. Cryptographic Evidence Trail & Cloud Datastore

- **Section 63 BSA Support:** Generates evidentiary metadata and audit logs supporting Section 63 BSA 2023 digital evidence workflows.
- **Merkle Chain of Custody:** Anchors source images, extracted tokens, and officer decisions in a tamper-evident SHA-256 hash tree.
- **Supabase Cloud Storage & PostgreSQL:** Provides persistent cloud object storage (`evidence-images` bucket) and PostgreSQL 16 datastore with complete cryptographic disposal linking.
- **Form-1 Statutory Notice PDF:** Compiles formal inspection memos with evidence crops, cryptographic digests, and signature blocks.

7. Offline Field Resilience (Mode B)

- **Local Standalone Runner:** Enables field officers to conduct inspections on standalone laptops (`localhost:8000`) with local SQLite.
- **Cryptographic Sync Bundles:** Packages offline inspection cases and evidence into signed bundles for central reconciliation.

4-State Uncertainty-Aware Verdict Model

Nirikshak classifies evaluated checks into four clear operational states:

Verdict State	Operational Definition & System Action
PASS	No configured statutory discrepancy detected across the evaluated declarations and measurements.
FAIL	Rule-engine discrepancy detected based on available evidence (e.g., font deficit, prohibited unit 'gms', or USP math mismatch > 0.02 INR).
REVIEW	Borderline measurement falling within the sensor uncertainty band (k=2, 95% confidence). Officer must conduct physical verification.
UNABLE_TO_VERIFY	Optical quality degraded (excessive blur, glare bloom, or occlusion). Directs officer to guided recapture flow before processing.

Real-World Validation & Proof

Nirikshak is backed by empirical testing on real retail commodities and automated verification:



Nirikshak real-world proof and verification metrics

Golden Demonstration SKUs

Thirteen representative reference commodities (7 statutory edge cases + 6 certified real physical packaging cases) are packaged directly into the workspace and catalog for instant demonstration:

SKU / ID	Scenario	Result	Key Diagnostic Observation & Legal Basis
DEMO-01	Sunfeast Butter Cookies 200g	FAIL	Table-I font deficit ($2.8\text{ mm} < 4.0\text{ mm}$); prohibited unit gms
DEMO-02	Everest Garam Masala 100g	FAIL	USP mismatch ($\text{Rs } 0.45/\text{g}$ declared vs $\text{Rs } 0.52/\text{g}$ computed)
DEMO-03	Himalayan Mineral Water 1L	PASS	Compliant baseline (numeral font 4.2 mm , valid unit L, correct USP)
DEMO-04	Medimix Ayurvedic Soap 75g	REVIEW	Sensor uncertainty (2.48 mm observed within $\pm 0.05\text{ mm}$ threshold band)

SKU / ID	Scenario	Result	Key Diagnostic Observation & Legal Basis
DEMO-05	Kurkure Masala Munch 90g	UNABLE_TO_VERIFY	Specular glare bloom obscuring net quantity panel; routes to recapture
DEMO-06	Royal Delight Almonds 500g	FAIL	E-Commerce: missing Country of Origin declaration; exempts Mfg Date
DEMO-07	Fortune Sunlite Oil 1L	FAIL	Indic Hindi Devanagari Net Qty numeral font deficit (2.9 mm $< 4.0\text{ mm}$)
REAL-PKG-WATCH	Fastrack Wristwatch (Item 1)	PASS	Single unit count (01 NUMBER $\rightarrow$ 1 N) exempt from USP under Rule 6(1)(da) Second Proviso; 180° inverted label auto-rectified
REAL-PKG-BRAHMI	Himalaya Brahmi 60T (Item 2)	PASS	Tri-panel fusion: Front PDP, Side LM panel with rate Rs. 4.33/TAB. , Back panel registered office Bangalore 560058
REAL-PKG-FACEWASH	Dot & Key Face Wash 100ml (Item 3)	PASS	Dual corporate entity decoupling (Mfg Solan HP vs Mkt Kolkata WB); volume rate ₹2.49/ml verified
REAL-PKG-PERFUME	Bella Vita Perfume 20ml (Item 4)	PASS	Base panel batch stamp fusion; volume rate ₹19.95/ml ; Gurugram Haryana manufacturing entity
REAL-PKG-NAMKEEN	Haldiram's Namkeen 400g (Item 5)	PASS	Clean SI unit symbol 400 g (0 prohibited gms flags); mass rate Rs. 0.25/g verified
REAL-PKG-CHIA	True Elements Chia 250g (Item 6)	PASS	Nutritional declaration isolation; rate ₹1.40/g ; Pune Maharashtra address & consumer care verified

System Architecture

Nirikshak is engineered as an **Online-First Modular Monolith** coupling a React 18 single-page application with an asynchronous FastAPI application service:

```
flowchart LR
    subgraph Client ["Client Tier (Browser)"]
        UI["React 18 SPA<br/>Vite 6 + Tailwind"]
        HUD["Camera Viewfinder HUD"]
        Canvas["Split-View Adjudication Canvas"]
    end

    subgraph Service ["Application Monolith"]
        API["FastAPI REST Catalog"]
        RBAC["JWT RBAC Engine"]
        Storage["Evidence Storage Manager"]
    end

    subgraph Pipeline ["12-Stage Diagnostic Pipeline"]
        direction TB
        QG["1. Quality Gate"]
        Calib["2. Scale Calibration"]
        OCR["3. Multilingual OCR"]
        Ext["4. Semantic Extraction"]
        Rule["5. Legal Rule Engine"]
        DAG["6. Cryptographic Provenance"]
        QG --> Calib --> OCR --> Ext --> Rule --> DAG
    end

    subgraph Output ["Evidentiary Outputs"]
        DB["PostgreSQL / SQLite"]
        Notice["ReportLab Form-1 Notice PDF"]
    end

    HUD --> UI
    Canvas <--> UI
    UI -->|HTTPS / REST| API
    API --> RBAC
    API --> Storage
    API --> Pipeline
    DAG --> DB
    DAG --> Notice
```

For comprehensive pipeline stage specifications, sequence diagrams, and mathematical formulations, see [docs/specifications/o3_FINAL_ARCHITECTURE.md](#).

Tech Stack

Layer	Technology	Operational Purpose
Frontend Workstation	React 18.3, Vite 6.0, TypeScript 5.7	Responsive inspector workspace, interactive adjudication canvas

Layer	Technology	Operational Purpose
Styling & UI Tokens	Tailwind CSS 3.4, Lucide React, Framer Motion	Government of India design tokens, accessible UI, micro-interactions
API & Backend	FastAPI 0.110+, Python 3.11+, Pydantic v2	High-throughput asynchronous REST catalog, typed schemas
Application Server	Uvicorn ASGI	Production web process server
Primary Datastore	PostgreSQL 16+	Relational schema, indices, JSONB evidence graphs
Local Field Datastore	SQLite 3.45+	Mode B offline field database for connectivity blackouts
Computer Vision	OpenCV 4.9+ (<code>opencv-python-headless</code>)	Laplacian blur variance, specular glare masking, ArUco fiducial calibration
Multilingual OCR	DBNet++ (Detection), PP-OCRv4 (Recognition)	ONNX Runtime CPU inference with quantized model variants for supported hardware
Fallback OCR	Tesseract 5 / pytesseract	Secondary fallback recognition engine
Compliance Rule Engine	Deterministic Python AST Engine	Codified statutory logic without probabilistic drift
Dossier & Notice Engine	ReportLab 4.1+	Cryptographically anchored Form-1 Statutory Notice PDF generator
Cryptography	<code>hashlib</code> (SHA-256)	Append-only Merkle DAG hash tree
Licensing	Apache-2.0, MIT, BSD-3-Clause	Apache-2.0, MIT, BSD-3-Clause — dependency stack reviewed for license compatibility.

Compact Project Structure

```
.
├─ main.py                # Monolith production server (FastAPI + SPA mount)
├─ local_runner.py        # Mode B standalone offline runner (localhost:8000)
├─ inspect_cli.py         # Headless CLI inspection utility
├─ requirements.txt       # Root Python dependencies
└─
```

```
|— backend/                                # CV, OCR, extraction, rules, evidence subsystems
|— contracts/                             # Frozen interfaces and schemas
|— integration/                           # Cross-subsystem fixtures and tests
|— frontend/                             # Inspector workstation frontend (React 18 + Vite)
|— docs/                                  # Technical specifications and guides
|— audit/                                 # Validation and ground-truth evidence
|— Legal Metrology real product images/# 75 Physical packaging test photos & collection notes
```

Getting Started

Prerequisites

- **Python:** Version 3.11, 3.12, or 3.13 (64-bit)
 - **Node.js:** Version 18.x or 20.x LTS with npm
 - **Database:** PostgreSQL 16+ (*Optional for local evaluation; automatic SQLite fallback is pre-configured*)
 - **Git:** Standard git client
-

Quickstart (Local Monolith)

1. Backend Setup

```
# Clone the repository
git clone https://github.com/anonymousgroupphp-collab/sih26034.git
cd sih26034

# Create and activate virtual environment
python -m venv .venv
# Windows PowerShell:
.venv\Scripts\Activate.ps1
# Linux / macOS:
source .venv/bin/activate

# Install dependencies
pip install --upgrade pip
pip install -r requirements.txt

# Start the application server
python main.py
```

The backend API will start at `http://localhost:8000`. Interactive OpenAPI documentation is accessible at `http://localhost:8000/docs`.

2. Frontend Setup

```
# In a new terminal, navigate to the frontend directory
cd frontend

# Install frontend packages
npm install

# Start Vite development server
npm run dev
```

Open your browser at `http://localhost:5173` . Requests to `/api` and `/storage` automatically proxy to `http://127.0.0.1:8000` .

3. Production Build

```
cd frontend
npm run build
```

The production build compiles into `frontend/dist` , which is served directly by `main.py` at `http://localhost:8000` .

Inspection Execution Modes

Standard Web Workflow

1. Open `http://localhost:5173` (or `http://localhost:8000`).
2. Log in using the seeded `Inspector` profile (quick-login buttons available in development mode).
3. Select **"New Inspection"** from the navigation bar.
4. Choose an intake mode: **Upload Photo**, **Live Camera Viewfinder**, **E-Commerce URL**, or **Demonstration Showcase**.
5. Execute the pipeline and review findings in the **Adjudication Canvas**.
6. Switch to the `Controller` role to review officer recommendations and generate the formal **Form-1 Statutory Notice PDF**.

Headless CLI Inspection

Inspect physical packaging directly from the command line:

```
python inspect_cli.py --image "Legal Metrology real product images/item 1 facewash/front_01.jpg" --re
```

Standalone Mode B (Offline Field Runner)

Execute inspections during field connectivity blackouts:

```
python local_runner.py
```

Verification & Automated Testing

Nirikshak enforces strict automated test verification across frontend, backend, and integration layers:

```
# 1. Run all Frontend Tests (142 passing tests across 18 test files)
cd frontend && npm test

# 2. Run Evidence & Backend + Integration Tests (95 passing tests)
python -m pytest backend/evidence/tests/ integration/tests/ -v

# 3. Run Computer Vision, OCR & Rule Engine Unit Tests (174 passing tests)
python -m pytest backend/cv/tests/ backend/ocr/tests/ backend/rule_engine/tests/ -v

# 4. Run Semantic Extraction Test Suite (151 passing tests)
python -m pytest backend/extraction/tests/ -v
```

Authoritative Test Summary

- **Frontend Test Suite (frontend):** 142 passing tests across 18 test files (0 failed, 0 skipped)
- **Backend & Subsystem Suites (backend/ & integration/):** 420 passing tests across 37 test files (0 failed, 0 skipped)
- **Platform Total:** 562 passing tests across 55 standard test files (0 failed, 0 skipped)

Safety, Workflow Safeguards & Evidentiary Support

1. **Augmented Assistant Boundary:** Nirikshak is strictly a diagnostic support tool. It possesses no autonomous legal authority and never issues notices or compounding orders independently.
2. **Mandatory Officer Adjudication:** Every automated finding is presented as a preliminary diagnostic observation. A qualified Legal Metrology Officer (LMO) must explicitly review, adjudicate, and sign off on each item.
3. **Mandatory Text Remarks on Override:** Any modification to an automated finding requires a mandatory written justification remark, preventing undocumented officer tampering.
4. **Separation of Duties (RBAC):** Field inspectors conduct intake and submit preliminary recommendations; only authorized district Controllers have administrative authority to dispose of cases or sign statutory notices.
5. **Section 63 BSA 2023 Evidentiary Workflows:** Electronic-evidence metadata and cryptographic hash manifests designed to support Section 63 Bharatiya Sakshya Adhiniyam, 2023 workflows, recording device parameters, software version digests, timestamps, and officer credentials.

6. **Pristine Evidence Preservation:** Ingested photographs are immutably preserved in storage (`/storage/uploads/`). Bounding polygons and coordinate annotations are maintained as distinct vector layers, ensuring original evidence pixels are never altered.

Operational Boundaries & Limitations

1. **Optical Image Quality:** Accurate sub-millimeter font height measurement requires photographs with adequate focus and resolution ($\geq 1080\text{p}$). Heavily degraded or blurry images are routed to `UNABLE_TO_VERIFY` for guided recapture.
 2. **Planar Reference Placement:** Physical scale calibration relies on planar homography. If a reference card or fiducial marker is placed on a different plane than the packaging face, perspective error may occur.
 3. **Hardware Execution Profile:** CPU inference with quantized model variants is designed for standard multi-core laptops. On low-power hardware, pipeline execution times may be longer than on standard workstations.
 4. **Human Adjudication Requirement:** Nirikshak does not substitute for authorized regulatory officers. Final legal adjudication, statutory notices, and enforcement actions remain the sole responsibility of human officials.
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Deep Dive Documentation

For comprehensive technical, legal, and operational specifications, consult the detailed documentation:

Domain	Canonical Reference	Focus Areas
System Architecture	docs/specifications/o3_FINAL_ARCHITECTURE.md	12-stage pipeline, subsystem interaction, data flows
Project Engineering Guide	docs/COMPLETE_PROJECT_END_TO_END_GUIDE.md	42-section comprehensive technical and domain guide
Specifications & ADRs	docs/specifications/	Frozen specifications (01 through 17), architectural

Domain	Canonical Reference	Focus Areas
		decision records
Technical Implementation	docs/technical/CAMERA_CAPTURE_IMPLEMENTATION.md	Viewfinder HUD, camera stream management, canvas controls
Research & Legal Dossiers	docs/research/	Statutory research dossiers, GSR Gazette analyses
UI/UX & Design System	docs/design/ & docs/ux/	Design tokens, wireframes, inspector user journey
Validation & Audits	audit/	Ground-truth SKU matrices, provenance registers, defect logs
Deployment Guide	docs/DEPLOYMENT_GUIDE.md	Docker Compose, reverse proxy configuration, cloud setup

Core Engineering Team

Developed for the **Department of Consumer Affairs (DoCA)** under **Smart India Hackathon 2026**:

Contributor	Workstream & Focus	Subsystem Scope	GitHub Profile
Kunal Raj	CV & Metrology	Optical quality gate, ArUco calibration, planar homography, PDP metric schedules	@kunal-raj-dev
Parmarth Kumar	Multilingual OCR & Lead Frontend	DBNet++ text detection, PP-OCRv4 Indic recognition, React 18 SPA, Adjudication Canvas	@parmarth-kumar
Harsh Patel	Semantic Extraction & Compliance Engine	Statutory entity parsing (MRP, Net Qty, Dates), banned unit detection, AST rule engine hardening	@anonymousgrouphp-collab
Ambika Bansal	Statutory Rule Engine	Table-I font schedules, USP math verification, temporal epoch routing, 4-state triage	@bansalambika12-ship-it
Shailendra Pratap Singh	Evidence & Backend	FastAPI REST services, PostgreSQL schema, Merkle DAG, Section 63 BSA support, Form-1 PDF	@shailendrapratap1
Urvashi Rajput	UI/UX Architecture	Workstation visual foundations, triage filter components, design tokens	@rajputurvashi2006-bit

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Department of Consumer Affairs (DoCA)

Ministry of Consumer Affairs, Food & Public Distribution • Government of India

Smart India Hackathon 2026 • Problem Statement ID: SIH26034